

HW0505_314707031

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```
library(AER)
library(car)
library(POE5Rdata)
library(knitr)
```

Chapter 11: Simultaneous Equations

Exercise 11.24: Fulton Fish Market

This exercise revisits the supply and demand model for fish at the Fulton Fish Market (Example 11.2), using the data file `fultonfish`. The structural demand equation (11.13) and supply equation (11.14) are

$$\ln(QUAN_t) = \alpha_1 + \alpha_2 \ln(PRICE_t) + \alpha_3 MON_t + \alpha_4 TUE_t + \alpha_5 WED_t + \alpha_6 THU_t + e_{dt}$$
$$\ln(QUAN_t) = \beta_1 + \beta_2 \ln(PRICE_t) + \beta_3 STORMY_t + e_{st}$$

In this data set `lprice` is $\ln(PRICE)$ and `lquan` is $\ln(QUAN)$. We add the indicator `mixed` (poor but not stormy weather) to the supply equation.

```
data("fultonfish")
```

11.24(a)

```
# Reduced form for ln(PRICE), equation (11.16) augmented with MIXED
rf_price <- lm(lprice ~ mon + tue + wed + thu + stormy + mixed,
              data = fultonfish)
summary(rf_price)
```

```
##
## Call:
## lm(formula = lprice ~ mon + tue + wed + thu + stormy + mixed,
##     data = fultonfish)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92281 -0.18819  0.00813  0.22404  0.68381
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -0.35957    0.07915   -4.543 1.50e-05 ***
## mon         -0.10825    0.10339   -1.047  0.29753
## tue         -0.06609    0.10103   -0.654  0.51446
## wed         -0.04927    0.10377   -0.475  0.63591
## thu          0.03935    0.10071    0.391  0.69681
## stormy       0.44634    0.07921    5.635 1.51e-07 ***
## mixed        0.23688    0.07851    3.017  0.00321 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3413 on 104 degrees of freedom
## Multiple R-squared:  0.245, Adjusted R-squared:  0.2014
## F-statistic: 5.624 on 6 and 104 DF, p-value: 4.349e-05
```

```
# Is MIXED significant at the 5% level?
```

```
coef(summary(rf_price))["mixed", ]
```

```
##      Estimate Std. Error    t value    Pr(>|t|)
## 0.236875284 0.078513425 3.017003561 0.003209729
```

```
# Joint significance of STORMY and MIXED
```

```
linearHypothesis(rf_price, c("stormy = 0", "mixed = 0"))
```

```
##
## Linear hypothesis test:
## stormy = 0
## mixed = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
##    Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     106 15.876
## 2     104 12.115  2     3.7604 16.14 7.857e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Explanation. The reduced-form equation for `lprice` now includes the two weather indicators `stormy` and `mixed` as the supply shift variables. The individual t-test shows whether `mixed` is significant at the 5% level (look at its p-value). The joint F test for `stormy = mixed = 0` is the relevance test for the instruments used to identify the demand equation. If the F-value exceeds 10, the instruments are strong enough that two-stage least squares of the demand equation can be trusted.

11.24(b)

```
# Demand equation (11.13) estimated by 2SLS using STORMY and MIXED as IVs
```

```
demand_iv <- ivreg(lquan ~ lprice + mon + tue + wed + thu |
                  mon + tue + wed + thu + stormy + mixed,
                  data = fultonfish)
summary(demand_iv)
```

```
##
## Call:
## ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
##       tue + wed + thu + stormy + mixed, data = fultonfish)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1327 -0.4320  0.0654  0.5133  1.1798
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.54023    0.15661  54.533 < 2e-16 ***
## lprice      -0.93014    0.35340  -2.632  0.00977 **
## mon         -0.01190    0.20837  -0.057  0.95456
## tue         -0.52583    0.20230  -2.599  0.01068 *
## wed         -0.56262    0.20696  -2.719  0.00767 **
## thu          0.09987    0.20285   0.492  0.62350
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6853 on 105 degrees of freedom
## Multiple R-Squared:  0.185,    Adjusted R-squared:  0.1462
## Wald test: 4.927 on 5 and 105 DF,  p-value: 0.0004317
```

Explanation. Here the demand equation is estimated by two-stage least squares, treating `lprice` as endogenous and using `stormy` and `mixed` as the excluded instruments. The coefficient on `lprice` is the price elasticity of demand, which is expected to be negative. Compared with Table 11.5 (which uses only `stormy`), adding `mixed` as a second instrument leaves the elasticity estimate close to the original value but can change its precision.

11.24(c)

```
# Sargan test of the overidentifying restriction (validity of the surplus IV)
summary(demand_iv, diagnostics = TRUE)$diagnostics
```

```
##              df1 df2  statistic      p-value
## Weak instruments    2 104 16.1400276 7.856950e-07
## Wu-Hausman         1 104  1.4891272 2.251117e-01
## Sargan              1  NA  0.7721695 3.795467e-01
```

Explanation. Because there are two instruments (`stormy`, `mixed`) for one endogenous variable (`lprice`), the demand equation is overidentified by one restriction, so the validity of the “surplus” instrument can be tested. The Sargan test regresses the 2SLS residuals on all instruments; under the null all instruments are valid. A large p-value (failure to reject) means there is no evidence that `mixed` is an invalid instrument.

11.24(d)

```
# In the reduced form from (a), test joint significance of the day indicators
linearHypothesis(rf_price,
  c("mon = 0", "tue = 0", "wed = 0", "thu = 0"))
```

```
##
## Linear hypothesis test:
## mon = 0
## tue = 0
## wed = 0
## thu = 0
##
## Model 1: restricted model
## Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed
##
##      Res.Df      RSS Df Sum of Sq      F Pr(>F)
## 1      108 12.408
## 2      104 12.115   4    0.29256 0.6279 0.6437
```

Explanation. To estimate the **supply** equation by 2SLS we need the demand-side shift variables — the daily indicators — to be jointly significant in the reduced form for `lprice`. The F test for `mon = tue = wed = thu = 0` checks this. As is typical with this data set, the day-of-week indicators are jointly insignificant, so they are weak instruments for `lprice`. Consequently we **cannot** estimate the supply equation by 2SLS with confidence, even though the demand equation can be estimated reliably.

Exercise 11.28: Truffle Supply and Demand with Price on the Left

This exercise uses the `truffles` data (Example 11.1). The original demand and supply equations (11.11) and (11.12) are

$$Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}, \quad Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$$

```
data("truffles")
```

11.28(a)

Rewriting the demand and supply equations with P on the left-hand side gives the inverse system

$$\text{Demand: } P_i = \alpha_1^* + \alpha_2^* Q_i + \alpha_3^* PS_i + \alpha_4^* DI_i + u_{di}$$

$$\text{Supply: } P_i = \beta_1^* + \beta_2^* Q_i + \beta_3^* PF_i + u_{si}$$

Anticipated signs. In the inverse **demand** equation the slope α_2^* should be **negative** (a downward-sloping demand curve), while α_3^* (price of the substitute PS) and α_4^* (income DI) should be **positive**, because both shift demand outward and raise price. In the inverse **supply** equation the slope β_2^* should be **positive** (an upward-sloping supply curve), while β_3^* (price of the factor PF) should be **negative**: a higher input cost shifts supply back, so at any given quantity a higher price is required, but holding Q fixed the coefficient on PF is expected to be negative.

11.28(b)

```
# Inverse demand by 2SLS: Q endogenous, IVs are the exogenous PS, DI, PF
inv_demand <- ivreg(p ~ q + ps + di | ps + di + pf, data = truffles)
summary(inv_demand)
```

```
##
## Call:
## ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -39.661  -6.781   2.410   8.320  20.251
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -11.428     13.592  -0.841  0.40810
## q             -2.671       1.175  -2.273  0.03154 *
## ps             3.461       1.116   3.103  0.00458 **
## di            13.390       2.747   4.875 4.68e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.17 on 26 degrees of freedom
## Multiple R-Squared: 0.5567, Adjusted R-squared: 0.5056
## Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06
```

```
# Inverse supply by 2SLS: Q endogenous, IVs are PS, DI, PF
inv_supply <- ivreg(p ~ q + pf | ps + di + pf, data = truffles)
summary(inv_supply)
```

```
##
## Call:
## ivreg(formula = p ~ q + pf | ps + di + pf, data = truffles)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.7983 -2.3440 -0.6281  2.4350 11.1600
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -58.7982     5.8592 -10.04 1.32e-10 ***
## q             2.9367     0.2158  13.61 1.32e-13 ***
## pf           2.9585     0.1560  18.97 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.399 on 27 degrees of freedom
## Multiple R-Squared: 0.9486, Adjusted R-squared: 0.9448
## Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16
```

Explanation. Both inverse equations are estimated by two-stage least squares, treating quantity q as the endogenous regressor and using the exogenous variables ps , di , and pf as instruments. The estimated slope on q is negative in the demand equation and positive in the supply equation, consistent with downward-sloping demand and upward-sloping supply, and the coefficients are generally significantly different from zero.

11.28(c)

```
# Price elasticity of demand "at the means" from the inverse demand estimates
a2 <- coef(inv_demand)["q"]      # dP/dQ along the inverse demand curve
Pbar <- mean(truffles$p)
Qbar <- mean(truffles$q)

# elasticity of demand = (dQ/dP)*(P/Q) = (1/a2)*(Pbar/Qbar)
elas_demand <- (1 / a2) * (Pbar / Qbar)
cat("dP/dQ (slope) =", a2, "\n")
```

```
## dP/dQ (slope) = -2.670519
```

```
cat("P-bar =", Pbar, " Q-bar =", Qbar, "\n")
```

```
## P-bar = 62.724   Q-bar = 18.45833
```

```
cat("Price elasticity of demand at the means =", elas_demand, "\n")
```

```
## Price elasticity of demand at the means = -1.272464
```

Explanation. The inverse demand equation gives $dP/dQ = \hat{\alpha}_2^*$, so $dQ/dP = 1/\hat{\alpha}_2^*$. The price elasticity of demand at the means is $(dQ/dP) \times (\bar{P}/\bar{Q})$. The result is about -1.27 , i.e. negative as expected and (in absolute value) greater than one, indicating that truffle demand is price elastic at the sample means.

11.28(d)

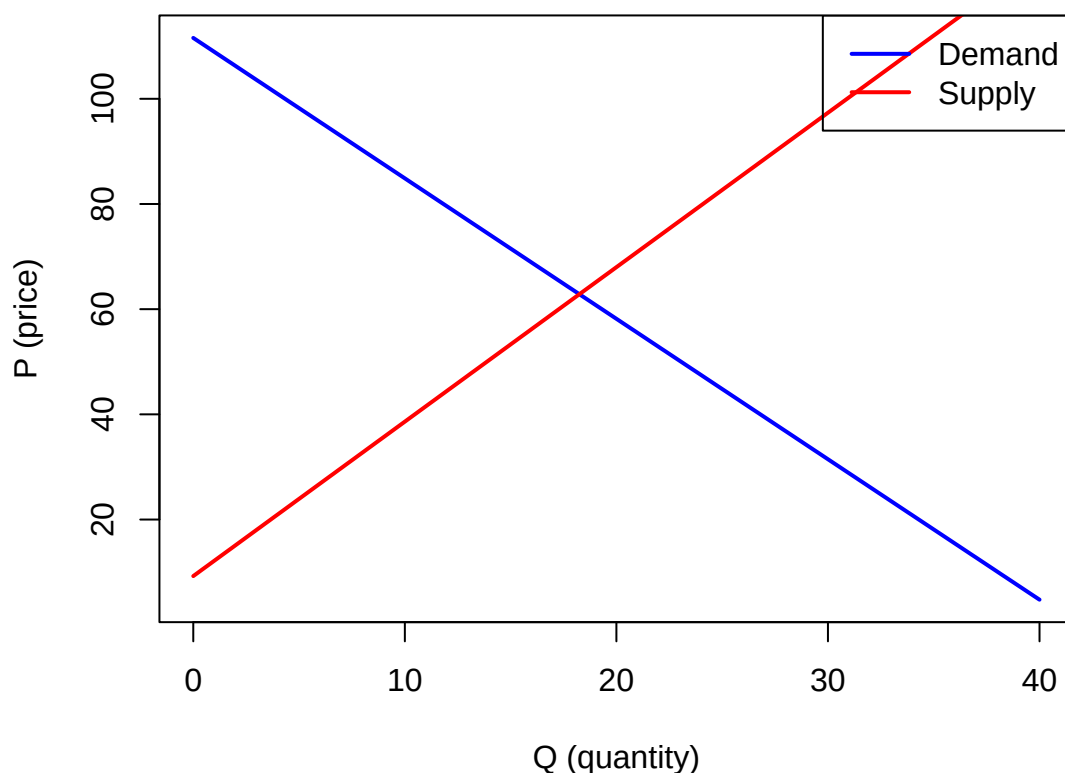
```
# Evaluate at the given exogenous values
DI_star <- 3.5; PF_star <- 23; PS_star <- 22

bd <- coef(inv_demand)    # inverse demand coefficients
bs <- coef(inv_supply)    # inverse supply coefficients

# P as a function of Q for each curve, holding exogenous variables fixed
demand_P <- function(Q) bd["(Intercept)"] + bd["q"]*Q + bd["ps"]*PS_star + bd["di"]*DI_star
supply_P <- function(Q) bs["(Intercept)"] + bs["q"]*Q + bs["pf"]*PF_star

Qseq <- seq(0, 40, length.out = 200)
plot(Qseq, demand_P(Qseq), type = "l", col = "blue", lwd = 2,
     xlab = "Q (quantity)", ylab = "P (price)",
     main = "Truffle Supply and Demand (P on vertical axis)")
lines(Qseq, supply_P(Qseq), col = "red", lwd = 2)
legend("topright", legend = c("Demand", "Supply"),
     col = c("blue", "red"), lwd = 2)
```

Truffle Supply and Demand (P on vertical axis)



Explanation. The curves are drawn with P on the vertical axis and Q on the horizontal axis, using $DI^* = 3.5$, $PF^* = 23$, $PS^* = 22$. The demand curve slopes downward and the supply curve slopes upward, intersecting at the market equilibrium.

11.28(e)

```
# Equilibrium from part (d): set demand_P(Q) = supply_P(Q) and solve for Q
# (bd0 + bd_q*Q) = (bs0 + bs_q*Q) with intercepts already including exog terms
bd0 <- bd["(Intercept)"] + bd["ps"]*PS_star + bd["di"]*DI_star
bs0 <- bs["(Intercept)"] + bs["pf"]*PF_star

Q_eq <- (bs0 - bd0) / (bd["q"] - bs["q"])
P_eq <- demand_P(Q_eq)
cat("Equilibrium from structural curves:  Q =", Q_eq, " P =", P_eq, "\n")
```

```
## Equilibrium from structural curves:  Q = 18.25021  P = 62.84257
```

```
# Compare with reduced-form predictions (Table 11.2): regress P and Q on all exogenous
rf_Q <- lm(q ~ ps + di + pf, data = truffles)
rf_P <- lm(p ~ ps + di + pf, data = truffles)
```

```
newx <- data.frame(ps = PS_star, di = DI_star, pf = PF_star)
Q_rf <- predict(rf_Q, newx)
P_rf <- predict(rf_P, newx)
cat("Equilibrium from reduced forms:      Q =", Q_rf, " P =", P_rf, "\n")
```

```
## Equilibrium from reduced forms:      Q = 18.2604  P = 62.81537
```

Explanation. Setting the two estimated inverse curves equal gives the equilibrium (Q, P) implied by the structural 2SLS estimates. The reduced-form equations predict P and Q directly from the exogenous variables at the same values DI^*, PF^*, PS^* . The two sets of equilibrium values are very close, confirming the internal consistency between the structural and reduced-form representations.

11.28(f)

```
# OLS of the inverse demand and supply equations
ols_demand <- lm(p ~ q + ps + di, data = truffles)
ols_supply <- lm(p ~ q + pf, data = truffles)
summary(ols_demand)
```

```
##
## Call:
## lm(formula = p ~ q + ps + di, data = truffles)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -25.0753  -2.7742  -0.4097   4.7079  17.4979
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -13.6195     9.0872  -1.499   0.1460
## q              0.1512     0.4988   0.303   0.7642
## ps             1.3607     0.5940   2.291   0.0303 *
## di            12.3582     1.8254   6.770 3.48e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.814 on 26 degrees of freedom
## Multiple R-squared:  0.8013, Adjusted R-squared:  0.7784
## F-statistic: 34.95 on 3 and 26 DF,  p-value: 2.842e-09
```

```
summary(ols_supply)
```

```
##
## Call:
## lm(formula = p ~ q + pf, data = truffles)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.4721  -3.3287   0.1861   2.0785  10.7513
```



```
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -52.8763     5.0238  -10.53 4.68e-11 ***
## q            2.6613     0.1712   15.54 5.42e-15 ***
## pf           2.9217     0.1482   19.71 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.202 on 27 degrees of freedom
## Multiple R-squared:  0.9531, Adjusted R-squared:  0.9496
## F-statistic: 274.4 on 2 and 27 DF,  p-value: < 2.2e-16
```

Explanation. Estimating the inverse equations by OLS ignores the simultaneity between P and Q , so q is correlated with the error term and the OLS estimates are biased and inconsistent. The signs are broadly the same as the 2SLS estimates, but the magnitude of the slope on q differs because OLS does not correct for endogeneity. The 2SLS estimates in part (b) are the preferred ones.